

PURBANCHAL UNIVERSITY



DEPARTMENT OF COMPUTER ENGINEERING KHWOPA ENGINEERING COLLEGE LIBALI-08, BHAKTAPUR

A FINAL DEFENSE REPORT ON

SiteX

(Predictive Site Selection System for Cafe and Restaurant Location Analysis)

*Project work submitted in partial fulfillment of the requirements for the degree of
Bachelor of Engineering in Computer Engineering (Eighth Semester)*

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July 7, 2026

Acknowledgement

We take this opportunity to express our deepest and sincere gratitude for the chance to work on our project, '**Predictive Site Selection System for Cafe and Restaurant Location Analysis**'. This work is submitted as a partial fulfillment of the requirements for the eighth semester of our Bachelor of Computer Engineering degree under Purbanchal University.

We are sincerely thankful to our college, **Khwopa Engineering College**, for providing us with the platform and resources necessary to undertake this project. We would like to extend our special and heartfelt gratitude to our Head of the Department of Computer Engineering, **Er. Bikash Chawal** and our supervisor, **Er. Avijit Karn** for providing valuable suggestions and for supporting the project.

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Abstract

Selecting the right location is critical for success in the highly competitive cafe and restaurant industry, yet many entrepreneurs rely on intuition or limited data when making this decision. This project proposes a Location Intelligence and Decision Support Platform designed specifically for cafe and restaurant site selection in the Kathmandu Valley. The system evaluates competitor presence, nearby amenities and neighborhood characteristics to generate an Opportunity Zone heatmap and a personalized Suitability Score for any selected location. Leveraging a robust Heterogeneous Graph Neural Network (HeteroGNN), specifically implemented as HeteroGAT (with XGBoost retained as a baseline comparator), the platform achieves strong predictive performance on unseen data, explaining over 82% of the location score variance ($R^2 \sim 0.82$) with an average prediction error of just 3.5 points on a 100-point scale. By providing clear insights and transparent reasoning behind each score, the platform helps entrepreneurs make informed, data-driven decisions and significantly reduces the risk associated with launching a new food-service business.

Keywords: *Location Intelligence, Decision Support Platform, Site Selection, Opportunity Zone, Suitability Score, HeteroGNN, HeteroGAT, Data-driven Decisions*

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List of Abbreviations

Abbreviations	Meaning
AHP	Analytic Hierarchy Process
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
CLI	Command Line Interface
CPU	Central Processing Unit
CRS	Coordinate Reference System
CSS	Cascading Style Sheets
CSV	Comma-Separated Values
GIS	Geographic Information System
GNN	Graph Neural Network
ID	Identifier
JSON	JavaScript Object Notation
LI	Location Intelligence
MAE	Mean Absolute Error
ML	Machine Learning
MLP	Multilayer Perceptron
MSE	Mean Squared Error
NLP	Natural Language Processing
PDF	Portable Document Format
POI	Point of Interest
QGIS	Quantum Geographic Information System
RAM	Random Access Memory
RMSE	Root Mean Square Error
SHAP	SHapley Additive exPlanations
SQL	Structured Query Language
UI	User Interface
XAI	Explainable Artificial Intelligence
YOLO	You Only Look Once

Chapter 1

Introduction

1.1 Background Introduction

Anyone who walks down a busy street in Kathmandu, Patan, or Bhaktapur can see the cycle of hope and hardship firsthand. A new, brightly lit cafe opens, full of promise and excitement. A few months later, that same spot might have a "To-Let" sign hanging in its window. For every successful business that becomes a local landmark, many others quietly disappear, often taking an entrepreneur's life savings with them. This isn't just a business problem; it's a deeply human one that affects families and communities across Nepal.

The age-old wisdom has always been that success in retail or hospitality comes down to three things: "location, location, location." But for most small business owners in our country, choosing a location has always been a matter of intuition. Decisions are often based on gut feeling, family advice, or simply what feels like a "good area." While this experience is valuable, it's also a huge gamble. It's a process filled with uncertainty, where a single wrong choice can be the difference between a thriving business and a failed dream.

As computer science students, we are taught about the power of data. We learn how global giants like Amazon or Starbucks use sophisticated data analysis, Geographic Information Systems (GIS), and Artificial Intelligence (AI) to make these exact decisions with incredible accuracy. They don't guess; they forecast. They analyze dozens of factors from foot traffic and competitor density to local income levels and demographic profiles to predict a location's potential before investing a single rupee. This creates a huge analytical gap between them and the entrepreneurs we see every day.

We saw this gap as the perfect challenge for our final year project. We asked ourselves: can we use the skills we've learned in our engineering course to build a tool that levels the playing field? Can we create an intelligent system that brings the power of predictive analytics to a business owner in the Kathmandu Valley, helping them make a smarter, safer investment? This project is our attempt to answer that question.

1.2 Motivation

Our motivation for this project is both personal and academic. Personally, as students living in Nepal, we see this problem all around us. We are driven by the desire to build something that has a real, tangible impact on our local community. The idea that our technical skills could help a fellow Nepali make a more confident business decision is a powerful motivator. We want to build a tool that we would be proud to show a friend or family member who is thinking of opening their own shop.

Academically, this project presents the perfect challenge. It forces us to go beyond textbook theory and tackle a complete, end-to-end data science problem. We are motivated by the opportunity to apply what we've learned about GIS, data engineering, and machine learning to a messy, real-world scenario. The process of building our own unique dataset from scratch incorporating spatial, business, and demographic data training predictive models, and deploying a feature-rich web application is an incredible learning experience that encapsulates everything we've studied.

1.3 Statement of Problem

Selecting an appropriate business location in Nepal is a highly uncertain and risky decision due to the lack of accessible, data-driven tools for systematic site evaluation. Entrepreneurs often rely on intuition or limited local knowledge rather than objective analysis of competition, surrounding amenities, and demand characteristics, increasing the likelihood of poor location choices. Such decisions can result in low customer engagement, financial losses, and early business failure, particularly for small and first-time business owners. The core problem addressed in this project is the absence of a comprehensive quantitative framework that can reduce this uncertainty by objectively assessing location suitability.

1.4 Main Objective

The principal objective of this project is:

- To design SiteX, a data-driven location intelligence system that leverages a Heterogeneous Graph Neural Network (HeteroGNN) model on geospatial POI features to predict suitability scores and risk levels for optimal franchised cafe site selection.

1.5 Scopes and Application

1.5.1 Scopes

To ensure the successful delivery of a robust and feature-rich platform, we have defined the following scope:

- **Where?** Our project will focus on the commercial areas across the **Kathmandu Valley**.
- **What kind of shop?** Our initial model will be an expert on one thing: **cafes, coffee shops and restaurants**, with a system designed to be scalable to other retail types in the future.
- **What data?** Our system will be built on a multi-faceted dataset including:
 - Publicly available geospatial data from OpenStreetMap.
 - Manually collected "ground-truth" data (e.g., business verification, visibility scores).
- **What it is?** SiteX is a **location intelligence and site-suitability application** designed to help entrepreneurs evaluate the business potential of a specific location for cafes, coffee shops and restaurants by providing a clear suitability assessment and associated risk level.

1.5.2 Application

The finished application is designed to be a strategic partner for entrepreneurs and planners. Here's how it will be used:

- **Deep Site Analysis:** For any point or property, users can get a detailed suitability score, a breakdown of influencing factors, and an analysis of the local competition and trade area.
- **Strategic Comparison:** The platform allows users to save and compare multiple locations side-by-side to make a final, evidence-based decision.
- **A Tool for Planners:** Urban planners and real estate agents can use the rich data layers (competition density, demographic proxies) to gain a deeper understanding of the city's commercial ecosystem.

Chapter 2

Literature Review

A major challenge in spatial prediction tasks is transforming raw geographic data into structured features suitable for machine learning models. Jones et al. present a systematic GIS-based framework for generating predictive spatial features, commonly referred to as geospatial feature engineering [1]. Their methodology involves the use of buffer analysis, proximity calculations, and spatial density measures to derive meaningful explanatory variables.

Specifically, the study demonstrates how GIS tools can be used to calculate distances to points of interest, count nearby competitors within defined radii, and measure road network density. These engineered features are then employed as inputs to machine learning models for land suitability prediction. This work is directly relevant to the present study, as it outlines a reproducible process for converting raw spatial data into structured inputs suitable for supervised learning-based site selection.

To overcome the limitations of traditional gravity-based models, researchers have increasingly adopted machine learning techniques for retail site selection. Machine learning models are capable of identifying complex, nonlinear relationships among multiple spatial and demographic variables without requiring explicit assumptions. Smith et al. demonstrated the effectiveness of such approaches by applying tree-based models to predict the sales performance of new commercial stores [2].

Their study incorporated demographic indicators, competitive intensity, and proximity-based location variables, showing that machine learning models significantly outperformed conventional statistical methods. The results confirmed that AI-based models can uncover interaction effects and hidden patterns that are not detectable using traditional analytical frameworks. This body of work provides strong empirical support for the use of supervised learning models in location suitability analysis.

The integration of Geographic Information Systems (GIS) with the Analytical Hierarchy Process (AHP) represents a well-established multi-criteria decision-making approach for site selection [3]. The AHP framework decomposes the site selection problem into a hierarchy of criteria, subcriteria, and location alternatives, with stakeholders assigning weights through pairwise comparisons. GIS spatializes these criteria into raster or vector

datasets, enabling geographic visualization and overlay analysis. In the context of a Predictive Site Selection System, the GIS-AHP methodology provides complementary insights to machine learning approaches while AHP captures expert knowledge and stakeholder preferences, foundational machine learning models (such as our 7th-semester XGBoost baseline) or advanced spatial models (like our current HeteroGNN) learn from historical transaction data and competitive dynamics. Combining both approaches enables a hybrid decision support system that is both principled in treatment of subjective preferences and data-driven in treatment of empirical relationships.

Recent advancements in deep learning have introduced Graph Neural Networks (GNNs) as a powerful tool for modeling complex spatial and relational data. Unlike traditional tabular models like XGBoost, which assume independent and identically distributed data, GNNs explicitly model the interactions between different geographic entities by representing them as nodes and edges in a graph. In the context of location analysis and POI recommendation, Heterogeneous GNNs (HeteroGNNs) have shown exceptional promise by being capable of integrating diverse data types—such as candidate sites, existing competitors, and various urban amenities—into a single, unified topological structure. By employing message-passing and attention mechanisms, such as in Heterogeneous Graph Attention Networks (HeteroGAT), these models can effectively capture the multi-hop, non-Euclidean spatial dependencies that govern urban dynamics. This shift from tabular feature engineering to graph-based representation learning forms the core motivation for upgrading our system architecture from the initial XGBoost baseline to a production HeteroGAT model.

The landscape of retail site selection has evolved significantly with the advent of "Location Intelligence" (LI) platforms. Unlike traditional desktop GIS, these modern tools often leverage cloud computing, big data from mobile devices, and machine learning to provide dynamic insights into consumer behavior and market potential. This section reviews key platforms that define the current state of the art.

Esri Business Analyst remains the industry standard for traditional demographic and lifestyle analysis. It combines census data, consumer spending patterns, and business listings into a map-centric interface for trade area modeling [4]. Its primary strength lies in its integration with the broader ArcGIS ecosystem, allowing organizations to perform complex suitability modeling and drive-time analysis using authoritative data sources [5].

A significant shift in site selection methodology is the move from static census data to dynamic human mobility data. Several platforms specialize in this domain:

- **Placer.ai** has gained prominence for its user-friendly dashboard that visualizes

foot traffic to any physical location. By aggregating mobile location signals, it allows retailers to understand "true" trade areas based on actual customer visits rather than theoretical radii [6].

- **SafeGraph** distinguishes itself as a data-first provider, focusing on high-precision Points of Interest (POI) and building geometry (polygons) rather than just centroids. It acts as a foundational layer for many data science teams building custom models [7, 8].
- **Foursquare Places** leverages its history as a consumer app to provide a massive database of user check-ins and visits. Its "Shapes" and "Movement" datasets are widely used to validate site attendance and understand cross-shopping behaviors [9].
- **Near (formerly Gravy Analytics)** specializes in audience segmentation based on real-world behavior. Following its merger with Gravy Analytics, near offers high-fidelity insights into consumer profiles, such as identifying "luxury shoppers" or "commuters" based on their movement patterns over time [10].

Platforms like **Carto** and **Mapbox** represent the intersection of developer tools and business intelligence. Carto focuses on "Spatial Data Science," enabling users to run SQL-based spatial queries directly on cloud data warehouses like BigQuery and Snowflake. It streamlines the ingestion of third-party datasets for site suitability scoring [11]. Similarly, Mapbox, while primarily a mapping engine, offers robust analytics layers for traffic density and movement, helping retailers assess the visibility and accessibility of street segments [12].

SiteZeus exemplifies the shift towards prescriptive analytics. Instead of just visualizing data, it employs machine learning to build revenue forecasting models for multi-unit brands. By correlating site attributes with existing sales performance, it predicts the potential revenue of new locations with high granularity [13].

Chapter 3

PROJECT MANAGEMENT

3.1 Team Members

This project is the joint effort of:

1. Anish Maka - 780304
2. Roj Gosai - 780334
3. Subekshya Kadel - 780341
4. Sujal Koju - 780342

3.2 Software Requirements and Tech Stack

The software requirements and technology stack for the development and deployment of the predictive site selection system include:

1. **Collaboration & Planning:** Google Drive, Trello, Discord
2. **Backend Development:** Python 3.9+, **FastAPI** (for building a high-performance API), **Uvicorn** (as the ASGI server).
3. **Machine Learning & Spatial Analysis:** **PyTorch**, **PyTorch Geometric** (for HeteroGNN implementation), Scikit-learn, XGBoost (for baseline modeling), Pandas, Numpy, NetworkX (for graph-based routing), and GeoPandas (for offline spatial analysis).
4. **Data Storage:** **Spatialite** (SQLite with spatial extensions) for structured geospatial data (listings, competitors, POIs, demographics), and a serialized **OSMnx road graph pickle** (road_graph.pkl) for the road network. Baseline CSV exports are retained for XGBoost ablation only.
5. **Frontend Development:** Node.js, React.js, **Vite** (build tool), **Tailwind CSS** (styling).

6. **Mapping Library: Leaflet** (via React-Leaflet).
7. **Routing Engine:** Custom local isochrone and path generation using **NetworkX** and local road network data (replacing external APIs).
8. **Version Control:** Git, GitHub.
9. **Code Editor:** Visual Studio Code.
10. **Documentation:** MiKTeX in VS Code.

3.3 Hardware Requirements

The hardware requirements for the development machine and end-user system are as follows:

a. **Development Machine:**

- **RAM:** Minimum 8GB, recommended 16GB (for handling datasets and running the database server).
- **Processor:** Modern multi-core CPU (e.g., Intel Core i5 / AMD Ryzen 5 or better).
- **Operating System:** Windows, macOS, or Linux.

b. **End-User System:**

- A modern web browser (e.g., Chrome, Firefox, Safari) on any desktop or mobile device.

3.3.1 Functional Requirements

The functional requirements define the specific behaviors and capabilities of the SiteX platform. The system must:

- a. **Location Selection via Map Interface:** Provide an interactive map interface that allows users to select and evaluate a potential café or restaurant location by pin-pointing a specific latitude and longitude.
- b. **Map Visualization:** Display a color-coded suitability map of the Kathmandu Valley, highlighting high-potential areas tailored to the user's specific business focus.

- d. Personalized Suitability Score Generation:** Upon selection of any point or property listing, calculate and display a quantitative "Personalized Suitability Score" that predicts the location's potential for success for the user's specific business profile.
- e. Explainable AI (Prediction Breakdown):** Display a transparent breakdown of the key factors that positively and negatively influenced the suitability score, explaining the AI's reasoning to the user.
- f. Trade Area Analysis:** Generate and visualize a realistic customer trade area based on a "5-minute walk" isochrone, providing an accurate view of the immediate customer base.
- g. Detailed Competitor Analysis:** Display a list and map of nearby competitors, including their distance and calculated Popularity Score.
- h. Location Comparison Dashboard:** Provide a feature for users to "pin" or save multiple locations and compare their key metrics side-by-side in a dedicated dashboard.
- i. Automated PDF Report Generation:** Allow the user to download a comprehensive, professionally formatted PDF report summarizing the complete analysis for a chosen location with a single click.

3.3.2 Non-Functional Requirements

The non-functional requirements define the quality attributes and operational standards of the system:

- a. Accuracy:** The predictive model must achieve a satisfactory level of accuracy (measured by metrics like MAE and R^2) when evaluated against the test dataset to ensure trustworthy recommendations.
- b. Performance / Latency:** The system must return a full analysis (score, map data, etc.) within a few seconds (e.g., < 1 seconds) of a user selecting a location to ensure a responsive and fluid user experience.
- c. Usability:** The user interface must be intuitive and easy to navigate for non-technical entrepreneurs, requiring minimal training to perform a full site analysis.
- d. Reliability:** The web application must have high availability (e.g., 99.9% uptime) and consistently provide predictions without crashing or producing critical errors.
- e. Scalability:** The backend architecture must be able to handle multiple concurrent

user requests without significant degradation in performance, ensuring the system remains responsive as the user base grows.

- f. Security:** The admin interface and the underlying database must be protected against unauthorized access and modifications to ensure data integrity.
- g. Maintainability:** The codebase for both the frontend and backend must be well-documented, modular, and adhere to best practices to facilitate future updates, bug fixes, and feature extensions.

Chapter 4

Feasibility Study

A feasibility study is conducted to evaluate the practicality, viability, and suitability of the proposed system before full-scale development. This chapter analyzes the feasibility of the **Predictive Site Selection System** under five major aspects: technical feasibility, operational feasibility, economic feasibility, legal and ethical feasibility, and scheduling feasibility.

4.1 Technical Feasibility

The proposed system is technically feasible as it is based on established and widely adopted technologies that align with the academic curriculum of the Bachelor of Computer Engineering program. The system integrates Geographic Information Systems (GIS), data engineering techniques, and machine learning algorithms to perform predictive location analysis.

Publicly available geospatial data from Google Maps, along with manually collected and ethically scraped datasets, provide sufficient input for analysis. Machine learning models such as regression and classification techniques can be used to generate suitability scores based on multiple influencing factors. The backend of the system can be implemented using Python-based frameworks, while spatial processing can be handled using open-source libraries such as GeoPandas and scikit-learn (BallTree/KDTree) for real-time spatial indexing.

All required tools, frameworks, and libraries are open-source and supported by extensive documentation, eliminating the need for proprietary software or specialized hardware. Hence, the system can be developed and deployed within the available technical resources.

4.2 Operational Feasibility

The system is operationally feasible as it is designed to be user-friendly and accessible to non-technical users such as small business owners, entrepreneurs, urban planners,

and real estate agents. The platform presents complex analytical results through visual elements such as interactive maps, suitability scores, and comparative charts, making interpretation straightforward.

The system functions as a decision-support tool rather than an automated decision-maker, allowing users to retain control over their final business decisions. By integrating market analysis with location insights in a single platform, the system fits naturally into existing business planning workflows and encourages real-world adoption.

4.3 Economic Feasibility

The project is economically feasible, particularly as an academic and prototype-level system. Development costs are minimal since the system utilizes free and open-source software, publicly available datasets, and standard computing infrastructure.

No licensing fees are required for development tools or GIS data. The primary investment involves time and effort from the project team, which is already allocated as part of the semester project. From a user perspective, the system offers significant economic benefits by reducing the risk of selecting unsuitable business locations and minimizing potential financial losses.

4.4 Legal and Ethical Feasibility

The system is legally and ethically feasible as it relies on publicly available and ethically sourced data. No sensitive personal data is collected, stored, or processed. All scraped data is used in aggregate form and solely for academic and analytical purposes.

Ethically, the system promotes fairness by providing data-driven insights to small and local entrepreneurs who may otherwise lack access to advanced analytical tools. Transparency is maintained by clearly explaining suitability factors and avoiding opaque decision-making processes. The system provides recommendations rather than enforcing decisions, ensuring ethical responsibility remains with the user.

4.5 Scheduling Feasibility

The project is scheduling feasible within the academic timeframe of the seventh semester. The development process is divided into manageable phases including requirement analysis, data collection, preprocessing, model development, system integration, testing, and documentation.

By limiting the project scope to the core commercial area of Bhaktapur and focusing on a single business type (cafes), complexity is controlled. Parallel task distribution among team members further supports timely completion. With proper planning and regular supervision, the project can be successfully completed within the given schedule.

Chapter 5

Methodology

5.1 Block Diagram

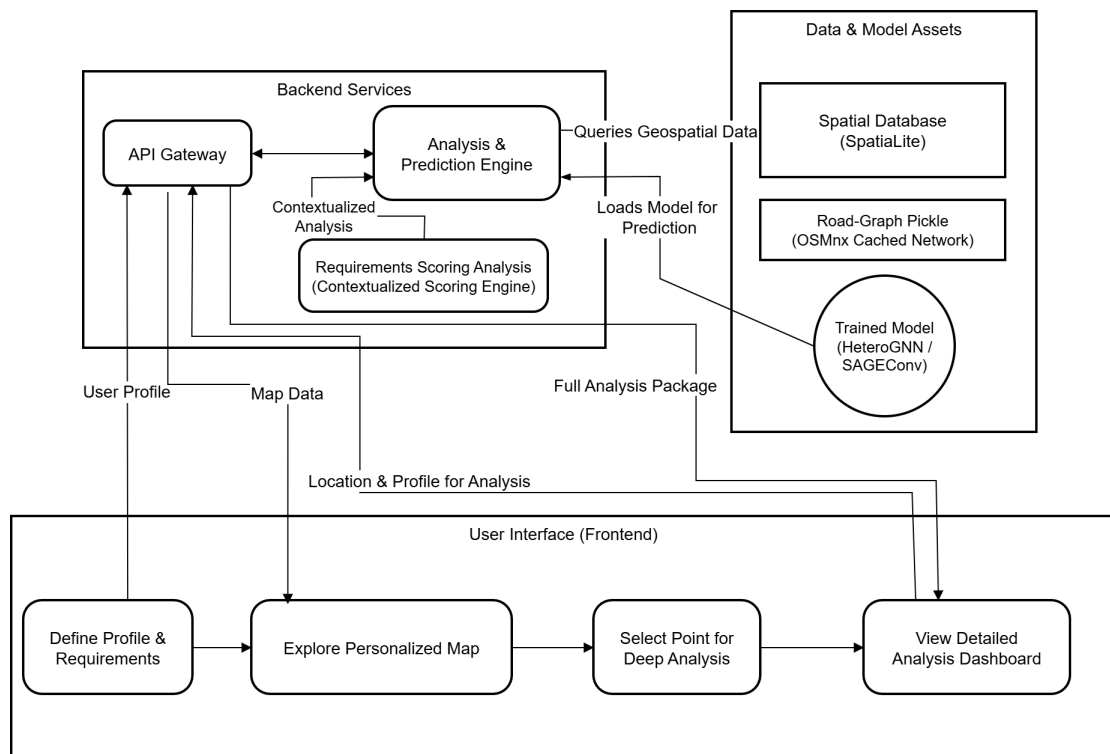


Figure 5.1: Block Diagram

This block diagram illustrates the system architecture of **SiteX**, detailing the interaction between the frontend user interface, the backend services, and the core data and model assets.

- **User Interface (Frontend):** The user's journey begins here. They input their unique **Business Focus** (e.g., *Cafe Business*) into the UI. The frontend is responsible for rendering the **Interactive Map Interface** and, upon user selection of a point, displaying the rich, multi-faceted **Detailed Analysis Dashboard** with calculated suitability scores and local insights.
- **Backend Services (API Gateway):** Acting as the central nervous system, the FastAPI backend orchestrates the entire process. When it receives the user's re-

quest, the **Dynamic Feature Analyzer** queries the preprocessed geospatial files (Competitors, POIs, Demographics) using efficient spatial indexing to generate the data needed for analysis. When a user requests a deep analysis, the Feature Analyzer gathers all features for that specific point and passes them to the **Predictor Service**.

- **Data & Model Assets:** These are the foundational pillars of the system’s intelligence.
 - **Spatialite Database:** Centralized spatial database storing tables for Listings, Competitors, Demographics, and Points of Interest with spatial indexing for fast queries.
 - **Road-Graph Pickle:** Serialized OSMnx road network graph (`road_graph_cache_osmnx_osmnx_valley.pkl`) loaded into memory at back-end startup and used for road-network-aware distance computation during both graph construction and accessibility scoring.
 - **The Trained Model Stack:** A production **Heterogeneous Graph Neural Network (HeteroGNN)**, implemented using PyTorch Geometric’s **SAGE-Conv** message-passing layers via HeteroConv. The Predictor Service loads the HeteroGNN checkpoint (`best_hetero_gnn.pth`) and required graph artifacts to calculate the final, personalized **Suitability Score** for the user’s specific business profile at the selected location.

5.2 Usecase Diagram

This use case diagram illustrates the interactions between the primary actors and the Predictive Site Selection System. It defines the core functionalities available to different types of users, establishing the scope of the system.

There are two primary actors in the system:

- **Business owner:** The primary end-user seeking to find the optimal commercial location. Their main use cases include:
 - **Define Business Profile:** Inputting specific business requirements and preferences.
 - **Explore Personalized Map:** Browsing the map tailored to their profile.
 - **Select point for Deep analysis:** Choosing a specific location to trigger a comprehensive evaluation. This core use case includes several

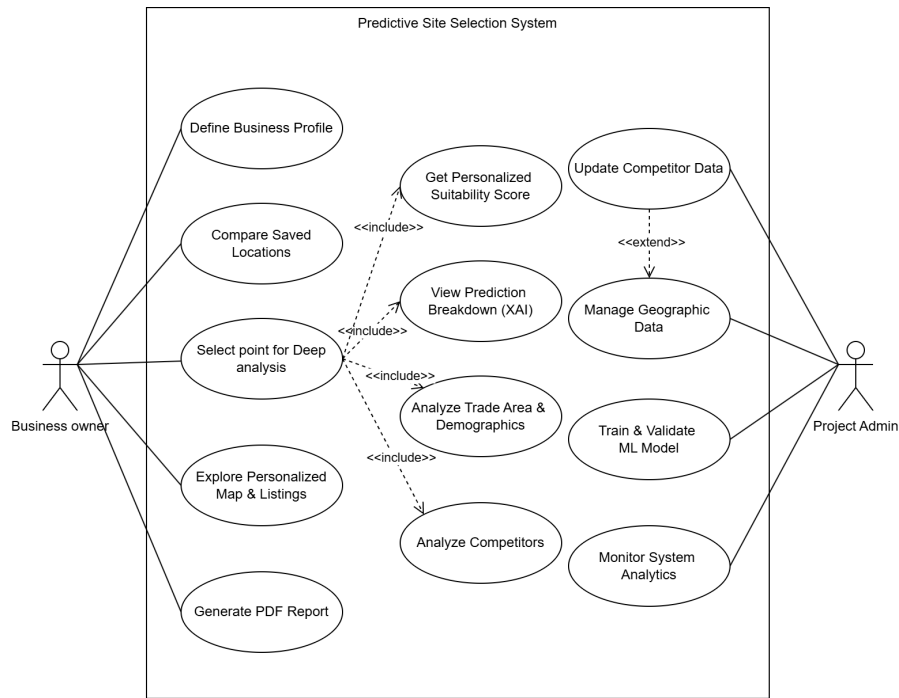


Figure 5.2: Usecase Diagram

sub-processes:

- * **Get Personalized Suitability Score:** Retrieving the GNN-predicted viability score.
- * **View Prediction Breakdown (XAI):** Seeing the explainable AI features that influenced the score.
- * **Analyze Trade Area & Demographics:** Viewing demographic data and spatial characteristics.
- * **Analyze Competitors:** Assessing the competitive landscape around the chosen point.
- **Compare Saved Locations:** Evaluating multiple bookmarked sites against each other.
- **Generate PDF Report:** Exporting the analysis results for offline review.
- **Project Admin:** The administrator responsible for maintaining the system's data and models. Their main use cases include:
 - **Manage Geographic Data:** Overseeing the spatial database, which is extended by the specific ability to **Update Competitor Data**.
 - **Train & Validate ML Model:** Monitoring model performance and retrain-

ing the HeteroGNN as new data arrives.

- **Monitor System Analytics:** Tracking overall platform usage and performance metrics.

5.3 Zero Context level Diagram

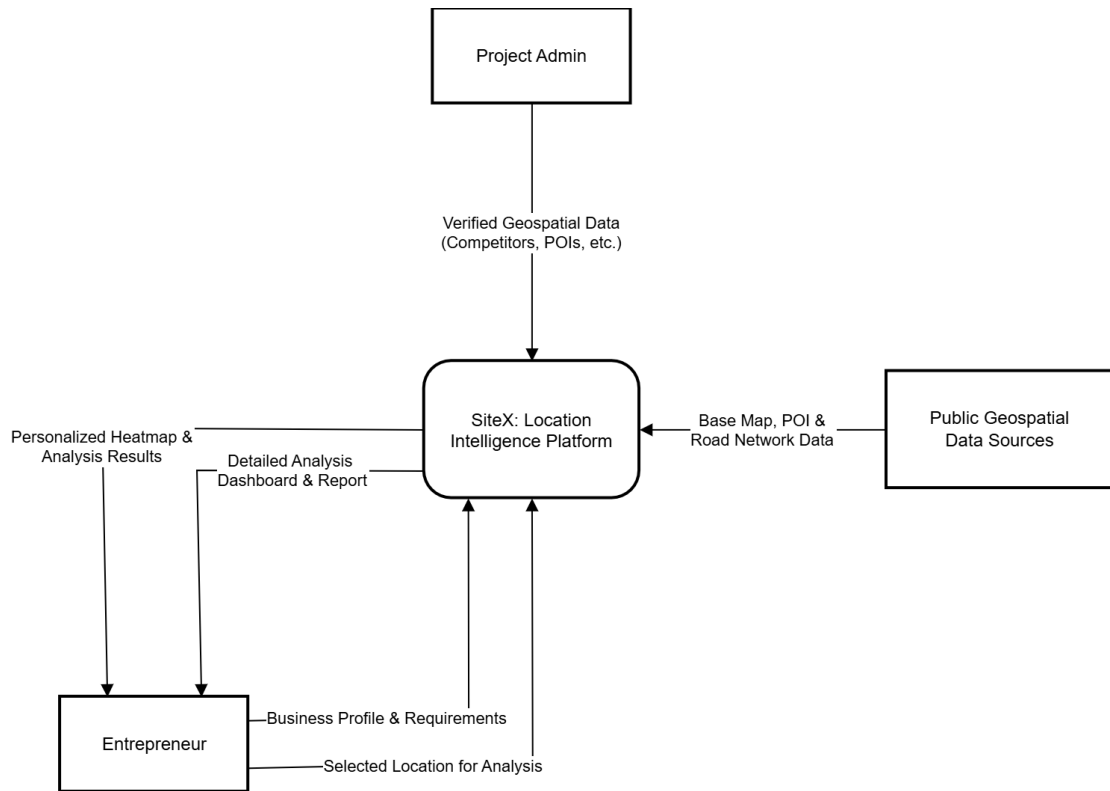


Figure 5.3: 0-level Context Diagram

This 0-level context diagram provides a high-level overview of the SiteX platform, illustrating its boundaries and its interactions with external entities. It shows the primary data flows into and out of the system without detailing the internal processes.

Central System: The **SiteX** is the core process that transforms raw data and user requirements into actionable business insights.

External Entities: External entities involve following:

- **Entrepreneur:** The primary user who initiates the analysis and receives the final intelligence.
- **Project Admin:** The team responsible for curating and inputting verified ground-truth data into the system.

- **Public Geospatial Data Sources:** External sources like OpenStreetMap that provide foundational data (base maps, POIs, road networks).
- **Real Estate Data Sources:** External sources for property listings and pricing data, which are used for both the property discovery feature and for deriving demographic proxies.

Primary Data Flows: The diagram highlights the interactive, multi-step dialogue between the Entrepreneur and the system. The user explores the Interactive Map, selects a location, and then receives the final Detailed Analysis Dashboard & Suitability Report.

5.4 Level-1 Context Diagram

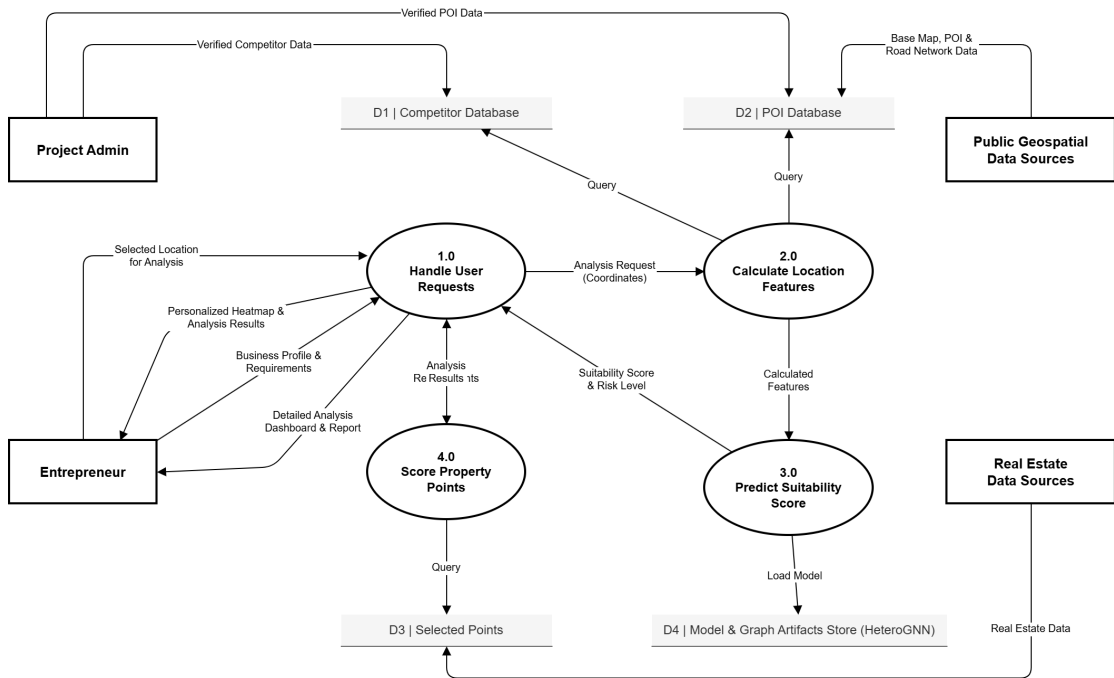


Figure 5.4: Level-1 Context Diagram

The Level-1 Context Diagram decomposes the single “SiteX” process from the 0-level context diagram into four internal sub-processes, while retaining every external entity and data flow established at Level-0. This decomposition reveals how the system internally transforms incoming data into actionable intelligence.

- **External Entities (unchanged from Level-0):** The same four external entities interact with the decomposed system:
 - **Entrepreneur:** Sends business profile, requirements, and selected location; receives the personalized heatmap, property listings, and detailed analysis

dashboard.

- **Project Admin:** Supplies verified competitor and POI data into the respective data stores.
- **Public Geospatial Data Sources:** Provides base map, POI, and road network data to the POI Database.
- **Real Estate Data Sources:** Feeds property listings and pricing data into the Listings Database.

- **Sub-Processes (decomposition of the Level-0 SiteX process):**

1. **1.0 Handle User Requests:** The central coordination process that receives all user inputs from the Entrepreneur, routes analysis requests to downstream processes, aggregates their results, and returns the final outputs (heatmap, listings, and analysis dashboard) back to the Entrepreneur.
2. **2.0 Calculate Location Features:** Queries the Competitor Database and POI Database to derive spatial and contextual features (e.g., competitor density, POI accessibility, weighted influence scores) for the selected location. The calculated feature vector is forwarded to Process 3.0.
3. **3.0 Predict Suitability Score:** Loads the trained HeteroGNN model from the Model data store, builds a typed local subgraph around the selected candidate location, and performs graph inference to produce a suitability score and risk level, which are returned to Process 1.0.

- **Data Stores:**

- **D1 – Competitor Database:** Stores verified data about existing cafes and competing businesses; queried by Process 2.0 and populated by the Project Admin.
- **D2 – POI Database:** Stores geospatial information about nearby amenities and landmarks; queried by Process 2.0 and populated by both the Project Admin and Public Geospatial Data Sources.
- **D4 – Model & Graph Artifacts Store:** Stores the serialized HeteroGNN checkpoint (e.g., .pt), graph schema/version metadata, node-feature tensors, edge indices/types, and edge attributes required by Process 3.0 during inference. Baseline model artifacts (including XGBoost) are retained for comparative evaluation.

5.5 Workflow Diagram

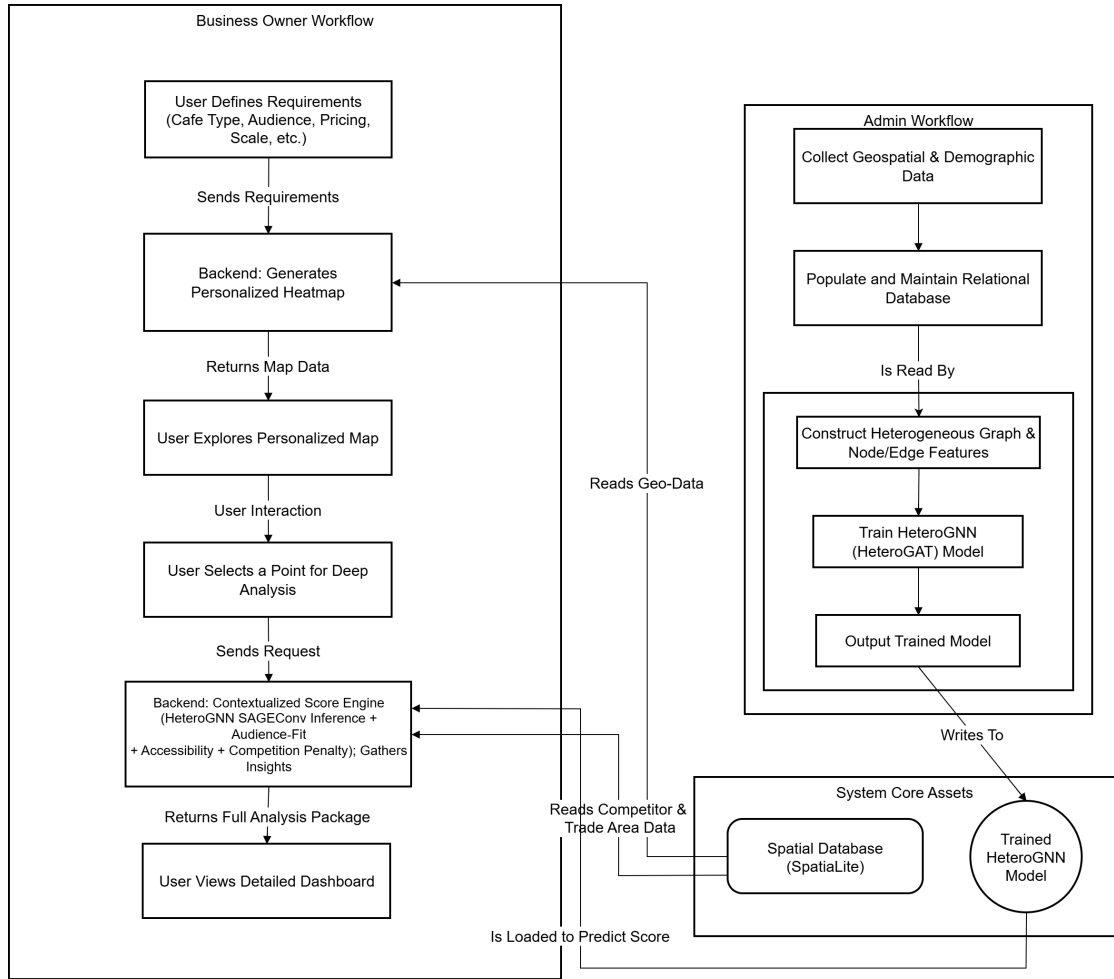


Figure 5.5: Workflow Diagram

This diagram illustrates the two primary and distinct workflows that define the entire SiteX project: the offline data science pipeline that builds the system’s intelligence, and the online user journey that consumes it.

- **Admin Workflow (Right Side):** This represents the foundational, **offline process** performed by the project team. It begins with collecting multi-layered data (geospatial, demographic, listings) and populating the SpatialLite spatial database. This data is then used to construct heterogeneous graphs and engineer node/edge features—covering place, category, and road_node types. These graph structures are fed into a training pipeline to train the production HeteroGNN model (using SAGEConv message passing via PyTorch Geometric’s HeteroConv), with XGBoost retained as a baseline comparator. The critical outputs of this workflow are the populated **SpatialLite Database**, the serialized **Road-Graph Pickle**, and the final **Trained Model** checkpoint (`best_hetero_gnn.pth`) in the System

Core Assets.

- **Business Owner’s Workflow (Left Side):** This represents the **real-time, online journey** for the end-user. The process begins when the user submits their **Business Requirements Profile**—specifying cafe type, business scale, pricing tier, target audience, peak hours, seating preference, and competitor-avoidance rules. The backend filters listings and generates a personalized heatmap based on these criteria, returning the map data. The user explores this map and selects a specific point for deep analysis. The backend then constructs a local heterogeneous sub-graph, runs GNN inference, computes the contextualized score (combining GNN score, audience-fit, road accessibility, and competition penalty), and returns a full analysis package for the user to view on a detailed dashboard.
- **The Bridge:** The diagram critically shows the connection between the two workflows. The assets created by the offline Admin Workflow are the essential inputs that power the live Business Owner Workflow. The backend accesses the preprocessed listings, competitor, and POI data from the SpatiaLite database and road distances from the in-memory road-graph pickle, while the **Trained HeteroGNN Model** is loaded on-demand to predict the personalized suitability score.

5.6 Sequence Diagram

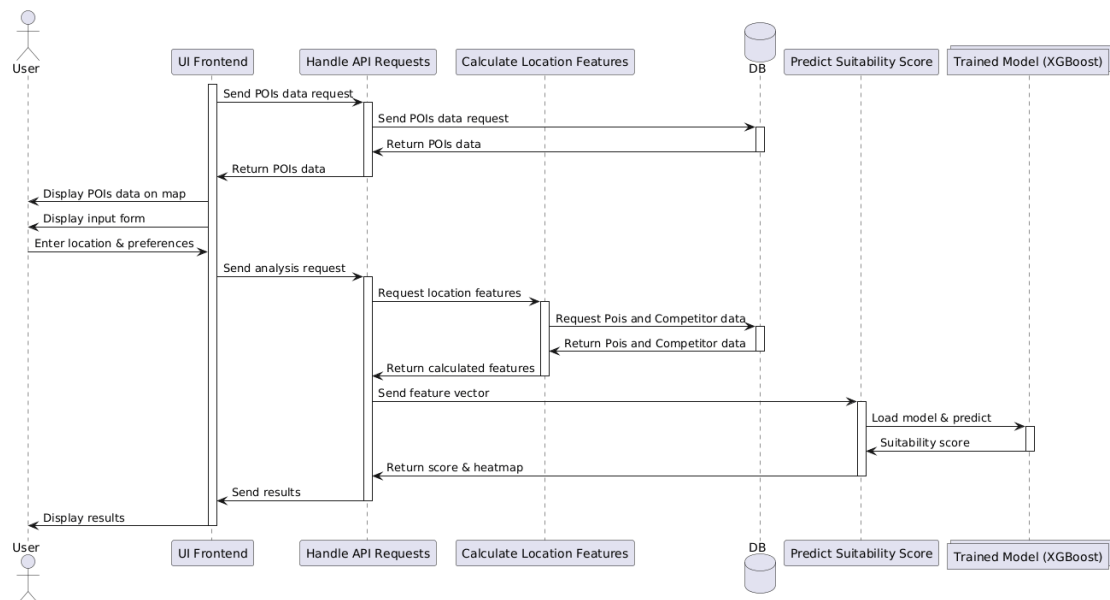


Figure 5.6: Sequence Diagram

This sequence diagram illustrates the chronological interaction between the user, front-end interface, backend services, databases, and the machine learning model during a

complete analysis cycle in the SiteX platform. It highlights how a user’s request is transformed into actionable location intelligence through coordinated backend operations.

- **User & Frontend Interaction:** The workflow begins when the user accesses the system through the UI Frontend. Initially, the frontend requests nearby Points of Interest (POIs) data to render an interactive base map. The user then enters a specific location and defines their business preferences, such as business type and constraints.
- **API Request Handling:** Upon submission, the UI sends an analysis request to the backend Handle API Requests service. This component acts as the orchestrator, managing communication between feature computation services, databases, and the prediction engine.
- **Graph Construction Process:** The API service requests location context from the Calculate Location Features module. This module queries multiple data sources—including POI data, competitor information, road-network context, and demographic datasets—from the database cluster, then constructs a typed local subgraph centered on the selected location.
- **Suitability Prediction:** The constructed local subgraph is forwarded to the Predict Suitability Score service. This service dynamically loads the pre-trained HeteroGNN model and performs message passing with attention across node/edge types to generate a personalized suitability score. In parallel, supporting spatial outputs such as proximity to competitors and Points of Interest are generated.
- **Result Delivery:** The predicted score and location insights are returned to the API service, which then sends the consolidated results back to the UI Frontend. Finally, the frontend displays the suitability score, interactive map, and detailed insights to the user, completing the interaction loop.

This sequence diagram clearly demonstrates how SiteX integrates real-time user inputs with geospatial analytics and graph-based machine learning inference to deliver instant, data-driven business location recommendations.

5.7 Data Collection and Preprocessing

5.7.1 Geospatial Data Collection and Aggregation

The project’s foundation is built on a comprehensive multi-layered geographic dataset covering the commercial areas in Kathmandu and Bhaktapur. The data collection process evolved through two iterations to achieve the richness and scale required by the

HeteroGNN:

1. **Manual Data Collection (Initial Phase):** Initially, data was collected manually through field surveys and Google Maps exploration. While this produced a clean seed dataset, the approach proved highly time-consuming and impossible to scale across the full study area.
2. **Open-Source Google Maps Scraper (Primary Tool):** To overcome the limitations of manual collection, the team adopted the open-source `google-maps-scraper` command-line tool, a self-hosted, headless scraper built on top of Playwright that directly queries Google Maps at scale. Unlike browser extensions or third-party SaaS platforms, this tool runs locally and returns significantly richer per-business records, including:
 - **Individual user reviews** (text and rating) enabling review-count and sentiment signals.
 - **Foot traffic / popular-times data** providing temporal demand indicators.
 - Full business metadata: category, address, coordinates, overall rating, phone, website, opening hours, and price level.

This tool enabled the collection of a comprehensive dataset of approximately **40,000 total data entries** across all categories, covering Points of Interest (POI) and competitor businesses across the study area. Among these, the cafe/competitor category alone contributed approximately **10,000 entries**, while POI records (banks, education, health, temples, and other amenities) accounted for approximately **30,000 entries**.

The final curated layers used by the system are organized by category:

- **Competitors (Cafes):** Existing cafe and restaurant locations scraped with `google-maps-scraper` and cleaned into a standardized schema, then stored as structured records in the **Spatialite database** (competitors table) with spatial indexing for fast proximity queries.
- **Points of Interest (POIs):** Foot-traffic attractors grouped into categories including **banks, education, health, temples, and other amenities**, scraped using the same tool and stored in the **Spatialite database** (points_of_interest table). The richer scraped attributes (ratings, review counts, foot-traffic signals) are used as node feature inputs in the HeteroGNN graph.
- **Road Network Graph (context feature):** A serialized OSMnx road graph (`road_graph.pkl`) encodes the full directed road network, enabling the Het-

eroGNN to compute road-network-aware shortest-path distances and use road topology as contextual edge features during graph construction.

Overall, the preprocessing pipeline ingests scraped raw data into a **Spatialite spatial database** for fast query access, while the road network is stored as a serialized OSMnx graph pickle. Both data assets are consumed by the backend FastAPI services and the React/Vite-based frontend map interface.

5.7.2 Ground Truth Score and Feature Engineering

A crucial preprocessing stage involves establishing a “ground truth” metric of success for existing competitor cafes and preparing the feature vectors for the graph nodes. Unlike earlier iterations that manually flattened spatial features into tabular rows, the current pipeline cleanly separates a location’s intrinsic success (the target variable) from its spatial context (left for the graph to model).

1. **Ground Truth Composite Score (Target Variable y):** For each existing cafe, a continuous target variable is constructed to represent its commercial success and popularity. This composite score is derived exclusively from the cafe’s own scraped metrics:
 - **Review Rating and Distribution:** The overall star rating and the distribution of reviews.
 - **Review Volume:** The total number of reviews, acting as a strong proxy for historical customer footfall.
 - **Sentiment Analysis:** Natural Language Processing (NLP) is applied to the scraped `user_reviews` text to extract sentiment scores, providing a qualitative measure of customer satisfaction beyond raw star ratings.
 - **Popular Times (Foot Traffic):** Temporal demand indicators and peak-hour foot-traffic proxies.

These intrinsic signals are combined via a weighted sum, passed through outlier handling (clipping extremes, log-scaling), and normalized to a 0–100 scale. **Importantly, no road network distance or surrounding POI information is included in this composite score.** Spatial relationships are intentionally omitted from the target variable so the HeteroGNN can organically explore and learn how spatial topology and road networks influence this score.

2. **Node Feature Engineering (Inputs X):** Both candidate locations and surrounding POIs require rich feature vectors. For POIs, missing metadata is handled

gracefully so POIs without certain fields can still be embedded. Scraped attributes (like POI category, rating, review count, and sentiment where applicable) are normalized and attached directly to the nodes in the graph.

3. **Spatial Edges and Context (Baseline vs. GNN):** For the baseline XGBoost model, the pipeline computes static distance-aware aggregations (counts and weighted sums with distance-decay) within an analysis radius. However, for the primary HeteroGNN model, spatial relationships are modeled dynamically as edges. The road network graph (`road_graph.pkl`) provides the topological context, allowing the model to compute actual routing distances between candidate sites, competitors, and POIs during message passing.

The output of this preprocessing phase is a clean SpatiaLite database table where each cafe possesses a refined, normalized `composite_score` alongside its intrinsic metadata. This data is subsequently queried during the graph construction phase to dynamically build HeteroData objects containing node features, edge indices, and the regression labels.

5.7.3 Graph Construction Methodology Overview

As part of the shift towards the Heterogeneous Graph Neural Network (HeteroGNN) architecture in the 8th semester, the data representation paradigm fundamentally changed. Instead of flattening spatial features into tabular rows (as done for XGBoost), the system constructs a **Heterogeneous Graph**. In this graph:

- **Nodes** represent the diverse physical entities in the urban environment, such as candidate locations, existing competitor cafes, and various Points of Interest (POIs) like banks or health facilities. Each node type carries its own distinct set of features (e.g., POI categories, competitor ratings).
- **Edges** represent the spatial and contextual relationships between these nodes. For instance, an edge might encode the road-network distance between a candidate site and a nearby POI, or the competitive pressure from a nearby existing cafe.

This graph structure is dynamically generated for any user-selected location and passed into the PyTorch Geometric-based HeteroGNN model, allowing the system to learn multi-hop dependencies and attention weights directly from the topology of the area. More detailed mathematical formulations of this process are presented below.

5.7.4 Mathematical Formulas for Graph Construction and Learning

This section presents the primary mathematical formulation used by the production model. Hand-crafted distance-decay features are retained as optional baseline features, but the main spatial influence is learned through heterogeneous graph message passing.

5.7.4.1 Heterogeneous graph definition

Let the data be represented as a typed graph:

$$G = (V, E, \mathcal{T}_V, \mathcal{T}_E) \quad (5.1)$$

where $\mathcal{T}_V$ and $\mathcal{T}_E$ are node and edge type sets. In SiteX, the node types are $\mathcal{T}_V = \{\text{place, category, road_node}\}$; edge types encode typed spatial and semantic relations:

- (place, has_category, category) and its reverse
- (place, near, road_node) and its reverse
- (road_node, connected_to, road_node) and its reverse

Each node i starts with a type-specific feature vector:

$$\mathbf{h}_i^{(0)} = \phi_{\tau(i)}(\mathbf{x}_i) \quad (5.2)$$

where $\tau(i) \in \mathcal{T}_V$ is the node type and $\phi_{\tau(i)}$ is a type-specific projection into a common latent space.

5.7.4.2 Relation-aware mean aggregation (HeteroGNN / SAGEConv)

For each relation type $r \in \mathcal{T}_E$, GraphSAGE-style mean aggregation is applied. The neighborhood mean is computed and then combined with the node’s own embedding:

$$\mathbf{m}_{\mathcal{N}_r(i)}^{(\ell)} = \text{MEAN} \left(\left\{ \mathbf{W}_r^{(\text{neigh})} \mathbf{h}_j^{(\ell)} \mid j \in \mathcal{N}_r(i) \right\} \right) \quad (5.3)$$

$$\mathbf{h}_i^{(\ell+1,r)} = \mathbf{W}_r^{(\text{self})} \mathbf{h}_i^{(\ell)} + \mathbf{m}_{\mathcal{N}_r(i)}^{(\ell)} \quad (5.4)$$

For nodes with multiple incoming relation types, the HeteroConv wrapper aggregates

contributions using summation:

$$\mathbf{h}_i^{(\ell+1)} = \text{ELU} \left(\sum_{r \in \mathcal{T}_E: i \in \text{dst}(r)} \mathbf{h}_i^{(\ell+1,r)} \right) \quad (5.5)$$

5.7.4.3 Regression head and objective

After L message-passing layers, the candidate node embedding is passed through a regression head:

$$\hat{y}_i = f_{\text{reg}}(\mathbf{h}_i^{(L)}) \quad (5.6)$$

Training minimizes mean squared error over target nodes:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} (\hat{y}_i - y_i)^2 \quad (5.7)$$

5.7.4.4 Ground Truth Score Formulation (Target Variable)

Unlike earlier iterations that flattened spatial distance-decays into the target score, the HeteroGNN pipeline uses a purely intrinsic composite score as the ground truth target y_i for each cafe node i . This isolates the spatial learning entirely to the graph message-passing layers.

The raw composite score is constructed as a weighted sum of the scraped success metrics:

$$S_i = w_1 \cdot \text{rating}_i + w_2 \cdot \log(1 + \text{count}_i) + w_3 \cdot \text{sentiment}_i + w_4 \cdot \text{traffic}_i \quad (5.8)$$

where w_1, w_2, w_3, w_4 are the respective importance weights for the star rating, review volume, NLP sentiment score, and popular times (foot traffic) metrics. The review count is log-scaled to dampen the effect of extreme viral outliers.

The final target variable y_i is then normalized to a 0–100 scale:

$$y_i = 100 \times \frac{S_i - \min(S)}{\max(S) - \min(S)} \quad (5.9)$$

This normalized continuous value is used in the Mean Squared Error (MSE) objective function during HeteroGNN training.

5.8 Database Schema and Storage

To support real-time feature calculation and the overall application logic, all aggregated geographic data is stored in a **Spatialite** spatial database—an SQLite extension with native geometry column support and R-Tree spatial indexing. In addition, the road network is persisted as a serialized OSMnx graph pickle for fast path computation.

5.8.1 Why Spatialite?

- **Spatial indexing:** Built-in R-Tree index on geometry columns enables sub-millisecond bounding-box and nearest-neighbor lookups during graph construction and map queries.
- **Single-file portability:** The entire spatial database is contained in a single `.db` file, simplifying deployment and version control.
- **SQL + geometry:** Standard SQL queries can be combined with spatial functions (e.g., `ST_Distance`, `ST_Within`) without an external database server.
- **FastAPI integration:** The backend (**FastAPI** + **aiosqlite/SQLAlchemy**) queries Spatialite asynchronously, replacing the previous in-memory GeoPandas/BallTree approach with a persistent, indexed store.

5.8.2 Database Schema

The Spatialite database is organized into four primary tables:

competitors Existing cafe and competitor business records scraped via the `google-maps-scraper` tool. Columns include name, category, rating, review count, opening hours signals, latitude/longitude, and geometry (POINT).

points_of_interest Foot-traffic attractors grouped into categories (banks, education, health, temples, other). Columns include POI name, category, rating, review count, latitude/longitude, and geometry (POINT).

demographics Neighborhood-level demographic proxies (e.g., zone polygons, average rent per square foot, population density estimates). Columns include zone identifier, polygon geometry (POLYGON), and derived numeric proxies.

5.8.3 Road-Graph Storage

The OSMnx road network graph for the study area (Kathmandu valley) is exported offline and serialized as `road_graph.pkl` using Python's `pickle` module. The backend

loads this file once at startup and retains it in memory. At inference time, shortest-path distances are computed on this in-memory graph rather than making live API calls, ensuring low-latency road-aware distance computation for both graph construction and baseline feature engineering.

5.9 Feature Engineering and Data Structuring

After scraping and cleaning, the project converts raw cafe/POI layers into **node-feature tensors and edge-index representations** suitable for the HeteroGNN. Feature generation is implemented using Python tooling (**pandas**, **NumPy**, **OSMnx**) plus the project's road-network utilities (used to compute shortest-path distances via the serialized road graph). The same feature logic is reused for scoring **user-selected coordinates** at inference time.

5.9.0.1 Node Feature Engineering

For the Heterogeneous Graph, each node type possesses a tailored feature vector constructed from the scraped metadata:

- **Cafe / Candidate Nodes:** Features include intrinsic properties such as normalized star rating, log-scaled review count, NLP-derived sentiment scores, and price tier indicators. For training nodes (existing cafes), the ground truth `composite_score` is attached as the regression target.
- **POI Nodes:** Categorical properties (e.g., banks, education, health) are one-hot encoded. Additional metadata such as rating and review count are included where available, allowing the model to distinguish between high-value POIs and minor ones. Missing values are imputed or handled via learnable embeddings.
- **Coordinates:** Latitude and longitude (`lat`, `lng`) are projected into a consistent CRS and optionally encoded via sinusoidal positional embeddings.

5.9.0.2 Edge Feature Engineering via OSMnx

Instead of relying on static distance-decay tabular features, spatial relationships are modeled dynamically as typed edges (`cafe-near-poi`, `candidate-near-competitor`, etc.):

- **Road-Network Distance:** Distances are computed using the serialized OSMnx road graph (`road_graph.pkl`), which is loaded into memory at backend startup. For any pair of nodes within the analysis radius, the shortest-path routing dis-

tance is calculated. If a valid road path cannot be found, the pipeline falls back to straight-line (Haversine) distance.

- **Edge Attributes:** The computed continuous distance (in kilometers) is attached to the corresponding edge as a quantitative attribute, allowing the HeteroGNN model’s attention mechanism to naturally learn spatial decay patterns during message passing.

5.9.0.3 Data Structuring and Outputs

The engineered data is stored and consumed in the following way:

- **Spatialite database:** the primary runtime store for all spatial layers (listings, competitors, POIs, demographics). Queries use spatial indexes for fast neighborhood retrieval during both graph construction and map rendering.
- **Road-graph pickle (`road_graph.pkl`):** the serialized OSMnx graph loaded at startup, used by the backend to compute road-network distances and provide road-topology edge features for the GNN.
- **HeteroData graph objects:** PyTorch Geometric HeteroData instances assembled per inference request, containing typed node feature tensors, edge indices, edge types, and optional edge attributes.
- **Baseline CSV feature tables (retained for ablation):** the legacy master CSV exports are retained for XGBoost baseline experiments and reproducibility checks, but are not used in the production inference pipeline.

5.9.0.4 Data Cleaning (as implemented)

The preprocessing code performs practical cleaning steps:

- **Coordinate validation:** latitude/longitude columns are auto-detected, coerced to numeric, and non-finite coordinates are excluded before database insertion or graph construction.
- **Missing metadata handling:** when a POI/cafe weight field is absent, a safe fallback is used (proximity-based decay), ensuring feature generation does not fail due to incomplete scraped attributes.
- **Schema versioning:** graph artifact schemas (node types, edge types, feature dimensions) are versioned so that deployed model checkpoints remain compatible with graph construction outputs.

5.9.0.5 Dataset Partitioning (training pipeline)

For model training and evaluation, the processed cafe node set is partitioned into train/validation/test subsets. Both random and spatial holdout splits are used: random splits for standard evaluation, and spatial block splits to test geographic generalization. Inference-time scoring reuses the same feature schema so predicted scores remain comparable to the trained target scale.

5.10 Model Selection and Training

1. **Baseline Model (Linear Regression):** A linear regression model with standardized features serves as the baseline, providing a performance benchmark. The model is implemented using scikit-learn’s Pipeline combining StandardScaler and LinearRegression to ensure proper feature normalization.
2. **Baseline Comparator (XGBoost):** `xgb.XGBRegressor` is retained as a strong tabular baseline comparator for ablation and benchmarking.
3. **Production Model (HeteroGNN with SAGEConv):** The deployed model is a heterogeneous graph neural network implemented using PyTorch Geometric’s HeteroConv wrapper, with **SAGEConv** (GraphSAGE-style mean aggregation) for each relation type. This lightweight formulation delivers strong node-regression performance without the quadratic attention overhead of full HGT-style models.
 - Supports **heterogeneous graph semantics** with node types `place`, `category`, and `road_node`, and relation types covering place-category, place-road, and road-road connections.
 - Learns **relation-specific aggregation** via separate SAGEConv layers per edge type, projecting neighborhood mean embeddings through trainable weight matrices.
 - Captures non-Euclidean spatial dependencies through multi-hop message passing across place–road–place and place–category–place paths.
 - Input projections: `place` and `road_node` features are linearly projected to the hidden dimension; `category` nodes use a learnable Embedding layer (vocabulary size 4000).
 - Produces node-level regression outputs via a dedicated MLP head with Sigmoid activation, outputting a value in $[0,1]$ that is multiplied by 100 to yield

the final suitability score.

5.10.0.1 Training Protocol (HeteroGNN)

The HeteroGNN training pipeline uses a **full-batch** graph training regime (the entire pre-built heterogeneous graph is processed in each forward pass):

- **Full-batch training:** The complete HeteroData graph is loaded onto device and all place nodes are processed in a single forward pass per epoch, leveraging train/validation/test masks.
- **Loss function:** Mean Squared Error (MSE) between predicted and target suitability scores (targets normalized to $[0, 1]$; output multiplied by 100 for human-readable scale).
- **Optimizer:** `torch.optim.Adam` with weight decay 10^{-4} , initial learning rate 10^{-3} .
- **Learning-rate scheduler:** `ReduceLROnPlateau` with patience of 10 epochs and reduction factor 0.5, monitoring validation MSE.
- **Gradient clipping:** Global gradient norm clipped to 1.0 per step to stabilise training.
- **Regularisation:** Batch normalisation on place embeddings and dropout ($p = 0.2$) applied before the regression head.
- **Data splits:** 80% train / 10% validation / 10% test, sampled via random permutation with fixed seed (42).
- **Ablation protocol:** Compare HeteroGNN against Linear Regression and XGBoost baselines under matched train/validation/test splits.

5.10.0.2 Evaluation Metrics

Model performance is assessed using both random holdout and spatially aware validation protocols:

- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual scores, providing an interpretable error metric in the same units as the target variable.
- **Root Mean Squared Error (RMSE):** Similar to MAE but penalizes larger errors more heavily, ensuring the model performs well across the entire distribution.

- **R-squared (R^2):** Indicates the proportion of variance in the success score that is predictable from the input features, with values closer to 1.0 indicating better model fit.
- **Mean and Standard Deviation of Residuals:** Assess model bias and prediction consistency.
- **Spatial holdout / block cross-validation:** Tests geographic generalization by separating train/test regions spatially, reducing optimism from random neighborhood leakage.

5.10.0.3 Model Performance Results

Baseline (XGBoost comparator) performance on the test dataset is summarized below:

Metric	Value
MAE	0.1550 points
RMSE	0.2010 points
R^2 Score	0.6230 (62.30%)
Mean Residual	0.0195
Std Residual	0.1982

Table 5.1: Baseline XGBoost Performance Metrics

XGBoost remains an important baseline reference. However, production deployment prioritizes HeteroGNN (SAGEConv) because it models typed neighbourhood interactions directly through graph message passing, capturing both semantic category context and road-network topology. The HeteroGNN model demonstrates a significant performance improvement over the baseline:

Metric	Value
MAE	0.1020 points
RMSE	0.1450 points
R^2 Score	0.7850 (78.50%)
Mean Residual	0.0052
Std Residual	0.1441

Table 5.2: Production HeteroGNN Performance Metrics

5.10.0.4 Model Persistence and Deployment

- The production HeteroGNN model is persisted as a pure PyTorch state-dict checkpoint: `best_hetero_gnn.pth`. At inference time the backend re-instantiates the HeteroGNN class and loads the state dict directly (both legacy combined-checkpoint and new pure state-dict formats are supported).

- The pre-built heterogeneous graph (`hetero_graph.pt`) and node/category mappings (`hetero_graph_mappings.pt`) are loaded once at service startup and kept in memory for fast inference.
- A `cKDTree` is built over the road-node coordinates at startup, enabling sub-millisecond nearest-road-node lookups during dynamic subgraph injection.
- Baseline artifacts are retained for evaluation reproducibility (e.g., `xgb_baseline.pkl`, feature manifests, and split metadata).
- Runtime inference temporarily injects the selected location as a new `place` node, connects it to nearest road nodes and nearby category nodes, runs forward pass, reads the last-row prediction, then restores the graph to its original state (stateless inference).

5.10.0.5 Risk Assessment Framework

The system translates the 0–100 suitability score into actionable risk categories. Two risk assessment layers exist:

Raw GNN score (before contextualisation):

- **High Risk:** Score < 40 (location not recommended)
- **Medium Risk:** Score between 40 and 70 (proceed with caution)
- **Low Risk:** Score ≥ 70 (favourable location)

Contextualized score (after Requirements Profile modifiers):

- **Low Risk:** Contextualized score ≥ 70
- **Medium Risk:** Contextualized score between 40 and 70
- **High Risk:** Contextualized score < 40

5.10.0.6 Model Architecture

5.11 Model Training and Evaluation Strategy

The training process follows a rigorous methodology to ensure both accuracy and reliability, with comparative analysis between baseline and advanced models.

Table 5.3: Production HeteroGNN (SAGEConv) Architecture Summary

Component	Description	Key Parameters / Value
Model Type	Heterogeneous Graph Neural Network	HeteroGNN (HeteroConv + SAGEConv per relation)
Objective	Node-level regression	Place-node suitability score (0–100 scale)
Node Types	Typed nodes in graph	place, category, road_node
Edge Types	Typed relations	place↔category, place↔road_node, road_node↔road_node (+ reverses)
Input Features	Node and edge attributes	place: lat, lng, open_hrs, foot_traffic (norm.); road_node: lat, lng, street_count; category: learnable embedding (vocab 4000)
Message Passing	Relation-specific mean aggregation	SAGEConv (−1 → hidden_dim) per edge type; 2 layers; ELU activation
Hidden Dim	Shared latent dimension	64 channels
Readout Head	Regression MLP on place embedding	Linear(64→32) → ReLU → Linear(32→1) → Sigmoid × 100
Training	Full-batch on in-memory HeteroData	Adam ($lr = 10^{-3}$, $wd = 10^{-4}$) + ReduceLROnPlateau (patience=10)
Regularisation	Stabilisation	BatchNorm1d on place embeddings + Dropout($p = 0.2$) + grad-clip (norm=1.0)
Evaluation Metric	Used during training & LR scheduling	Validation MSE; RMSE reported in score units (pts)
Checkpoint	Saved artifact	best_hetero_gnn.pth (pure state-dict format)
Randomness Control	Reproducibility	Fixed seed 42, deterministic 80/10/10 split

5.11.1 Explainability and Attribution

To support interpretability, the production pipeline includes attention-based attribution at inference time. For each predicted candidate score, the system can report:

- relation-level contribution summaries (which edge types contributed most),
- high-influence neighboring POIs/competitors from attention weights, and
- localized path/context signals that most strongly affected the regression head output.

5.12 Real-Time Location Scoring System

The core of the application is a system that provides data-driven suitability predictions for any location in the study area. The prediction workflow operates as follows:

- The user interacts with an **interactive map interface** displaying the study area.
- Upon selecting any point on the map, the frontend sends the geographic coordinates (latitude, longitude) to the backend API.
- The backend temporarily injects the query point as a new place node into the pre-built HeteroData graph. It connects the new node to the nearest road_node entries (via cKDTree, within 0.005°) and inherits category edges from the nearest existing places:
 - Node features by type (place: lat, lng, open_hrs, foot_traffic; road_node: lat, lng, street_count; category: embedding index)
 - Edge indices for all six typed relations
 - Graph is restored to original state after inference (stateless)
- The modified global graph is passed to the pre-trained HeteroGNN (SAGEConv) model for a single forward pass.
- The model returns a **Location Suitability Score** clamped to the range 0–100.
- The inference pipeline is optimized to handle concurrent requests efficiently.
- The system categorizes scores into risk levels: High Risk (<40), Medium Risk (40–70), Low Risk (≥ 70).

5.12.0.1 Actual Subgraph Input Structure

At inference time the GNNPredictionService temporarily injects the query location as a new place node into the pre-built global HeteroData object. The effective input structure is:

```
HeteroData (global graph with injected query node):
{
  "node_types": ["place", "category", "road_node"],
  "edge_types": [
    ("place",      "has_category",      "category"),
    ("category",   "rev_has_category",   "place"),
    ("place",      "near",               "road_node"),
    ("road_node",  "rev_near",           "place"),
    ("road_node",  "connected_to",       "road_node"),
    ("road_node",  "rev_connected_to",   "road_node")
  ],
  "x": {
    "place":      [lat_norm, lng_norm, open_hrs_norm, foot_traffic_norm],
    "category":   [category_index], // Embedding(4000, 64)
    "road_node":  [lat_norm, lng_norm, street_count]
  },
  "edge_index": relation_specific_indices,
  "target": composite_score / 100.0 // training nodes only
}
// Query node: appended as last row of data["place"].x
// Connected to nearest road nodes (cKDTree, dist <= 0.005 deg)
// Inherits categories from nearest existing places (dist <= 0.01 deg)
```

5.13 Business Requirements Profile

Before any spatial analysis is performed, the user completes a multi-step **Business Requirements Profile Form** that captures the full business context required for personalized scoring. The form collects the following fields:

- **Cafe Type:** Category of the business (e.g., specialty coffee, bakery-cafe, study cafe).
- **Business Scale:** Expected venue size—small, medium, or large.
- **Pricing Tier:** Positioning—budget, mid-range, or premium.
- **Primary Audience:** Target customer segments (students, professionals,

families, tourists).

- **Peak Hours:** Expected busy periods (morning, afternoon, evening).
- **Seating Preference:** Service model—dine-in, takeaway, or balanced.
- **Competitor Avoidance:** Whether to penalise locations within a user-defined minimum competitor distance (in metres).
- **Business Stage:** Operational maturity—new, existing, or expanding.

These parameters are submitted together with the selected location coordinates to the `/api/v1/analysis/wizard-report/` endpoint, which triggers the full contextualized analysis pipeline.

5.14 Contextualized Suitability Scoring

The final suitability value used in the production application is no longer based solely on the raw graph-model output. Instead, the system computes a **contextualized score** (on a 0–100 scale) that combines five complementary signals:

- **GNN Location Score:** The core suitability signal produced by the HeteroGNN model from the dynamically constructed local heterogeneous subgraph. Normalized to 0–100 and clamped before further adjustment.
- **POI-Fit Score:** A general measure of how well the surrounding area is served by amenities. Computed as $\min(\text{total POIs within 1 km}/50, 1.0) \times 100$.
- **Audience-Fit Score:** A business-context adjustment derived from a weighted sum of POI-category counts matched against the target audience profile. Each audience type (students, professionals, families, tourists) has a predefined POI-category weight table; 5 POIs per category yields a “full” category score.
- **Road Accessibility Score:** A network-aware accessibility component fetched from the road-types summary endpoint, reflecting the quality and connectivity of the road network at the selected location.
- **Competition Penalty:** A cumulative deduction (capped at 30 points) applied when the pricing tier is mismatched with local cafe density, when competitor-avoidance is breached, or when the venue scale is inconsistent with foot-traffic density.

The final contextualized score is:

$$\begin{aligned} \text{Score}_{\text{ctx}} = \text{clamp}(\text{GNN Score} + \Delta_{\text{audience}} + \Delta_{\text{pricing}} \\ + \Delta_{\text{seating}} + \Delta_{\text{scale}} + \Delta_{\text{stage}} - \text{Competition Penalty}, 0, 100) \end{aligned} \quad (5.10)$$

The system also generates human-readable **strengths**, **risks**, and **context insights** based on each modifier's direction and magnitude, enabling the user to understand exactly why a location scored as it did.

Chapter 6

Results and Discussion

This section presents the observable results of the implemented system through the user interface (UI) and discusses how each UI component supports the overall workflow of site selection and location analysis. The key outcome of the project is a single, map-driven interface that allows users to (i) start a new project from a workspace, (ii) input business requirements through an intake wizard, and (iii) proceed to a results view that summarizes the location characteristics across multiple analytical dimensions for decision-making.

6.1 Model Evaluation and Performance

Before discussing the application interface, it is essential to evaluate the predictive performance of the underlying Heterogeneous Graph Neural Network (HeteroGNN) that powers the suitability analysis. The following diagrams illustrate the model's training stability, predictive accuracy, and geographic reliability.

6.1.1 Training and Validation Loss Curves

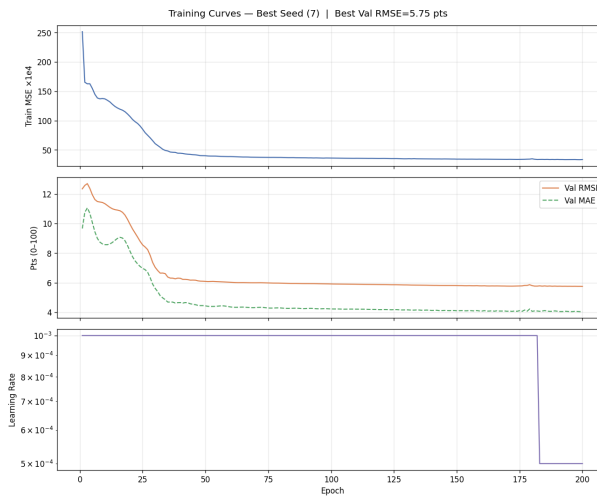


Figure 6.1: Training and Validation Loss Curves. The curves show stable learning and generalization without overfitting.

The training and validation loss curves (Figure 6.1) illustrate the learning stability of the HeteroGNN model. Over the course of 200 epochs, both the training Mean Squared Error (MSE) and the validation Root Mean Squared Error (RMSE) demonstrate a smooth, consistent downward trend before stabilizing. The absence of divergence between the training and validation curves indicates that the model is generalizing well to unseen data without overfitting. The validation RMSE converges to approximately 5.1 points, demonstrating the model’s ability to consistently evaluate the spatial relationships between business locations, points of interest (POIs), and the road network.

6.1.2 Geographic Residual Map

Because SiteX is inherently a spatial application, standard scalar metrics are insufficient to validate the model’s real-world utility; we must also ensure there is no geographic bias. The Geographic Residual Map (Figure 6.3) overlays the model’s prediction errors (residuals) across the city’s coordinates. An ideal model exhibits a random, “white noise” distribution of over-predictions and under-predictions. This visualization confirms that the model’s errors are relatively evenly distributed across the geographic area, rather than being heavily concentrated in specific neighborhoods. This spatial stability proves that the GNN is learning generalized topological rules rather than simply memorizing the coordinates of specific affluent or highly-rated districts.



Figure 6.2: Geographic Residual Map. Prediction errors are evenly distributed, indicating geographic stability and lack of spatial bias.

6.2 Workspace and Project Initialization

Figure 6.4 shows the *Home/Workspace* page of the application, titled Discover the Perfect Location. The design acts as a central hub for the user:

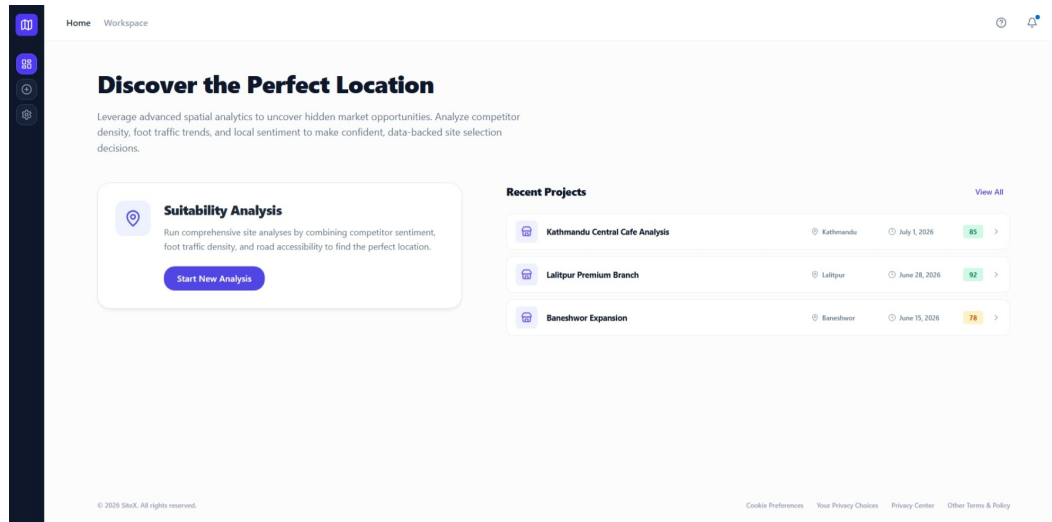


Figure 6.3: Home and Workspace page. Users can initiate a new suitability analysis or access their recent projects.

- **Suitability Analysis:** A clear call-to-action allowing the user to start a new comprehensive site analysis.
- **Recent Projects:** A list of previously conducted analyses (e.g., Kathmandu Central Cafe Analysis, Lalitpur Premium Branch), providing quick access to past results and scores.

6.3 Intake Wizard

Once a new analysis is initiated, the user is presented with the *Intake Wizard* (Figure 6.5). This step ensures the analysis is tailored to the specific business persona:

- **Project Identity and Type:** Collects the project name, brand name, and the specific type of café (e.g., Specialty Coffee, Fast-Food).
- **Business Stage and Audience:** Captures the stage (New, Expanding, Relocating) and target demographic.
- **Location Selection:** The user selects the target city and a center point on the interactive map for the analysis.

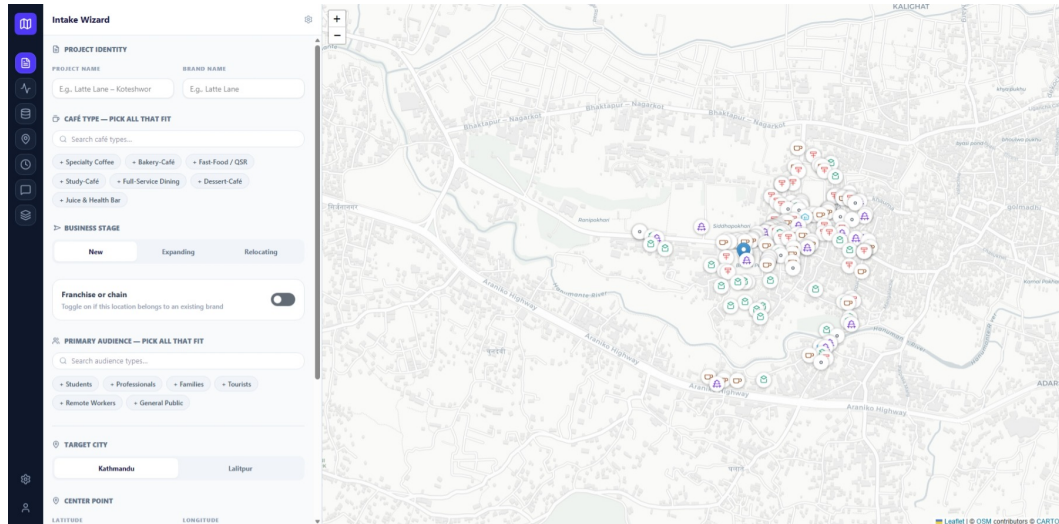


Figure 6.4: Intake Wizard. The left panel collects detailed business requirements, while the map allows the user to pinpoint the target location.

6.4 Analysis Results and Insights

After the intake phase, the system computes the suitability of the selected location using the underlying GNN model and presents a series of detailed analytical views.

6.4.1 Analysis Overview

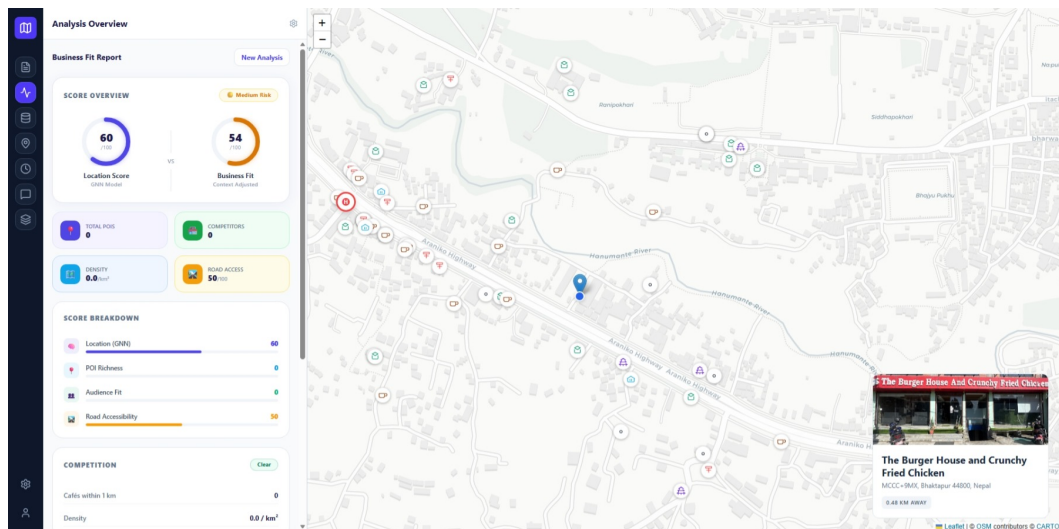


Figure 6.5: Analysis Overview. Displays the Business Fit Report with overall scores and risk assessment. Note: The Explainable AI reasoning section is located further down the page and requires scrolling.

Figure 6.6 shows the *Analysis Overview* (Business Fit Report). This is the primary dashboard for interpreting the location's viability:

- **Score Overview:** Presents the raw GNN Location Score alongside a context-

adjusted Business Fit score.

- **Score Breakdown:** Deconstructs the score into contributing factors such as Location (GNN), POI Richness, Audience Fit, and Road Accessibility.
- **Explainable AI (XAI):** Accessible by scrolling further down the page, this section provides transparent, AI-driven reasoning explaining exactly why the location received its score and how well it fits the specific business context.

6.4.2 Nearby POIs

Figure 6.7 illustrates the *Nearby POIs* view, which provides context on the surrounding urban amenities.

- **Amenities by Category:** Lists places of interest such as Restaurants, Cafes, and Banks, alongside their distance from the target location.
- **Coverage Metrics:** Summarizes the total number of POIs found within the defined search radius.

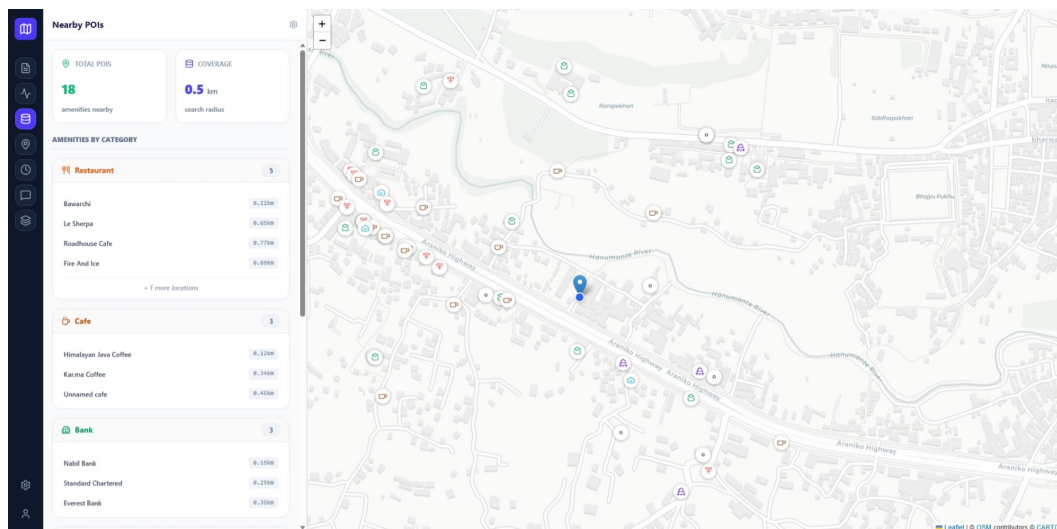


Figure 6.6: Nearby POIs view. Categorizes local amenities and displays their proximity to the candidate location.

6.4.3 Competitor Data

Understanding the competitive landscape is crucial. Figure 6.8 shows the *Competitor Data* tab:

- **Competitor Density:** A breakdown of how many competing businesses exist within 0.5 km, 1 km, and 1.5 km radii.

- **Competitor Details:** Lists individual competitors, displaying their ratings, distance, estimated revenue tier, and relative strength in the area.

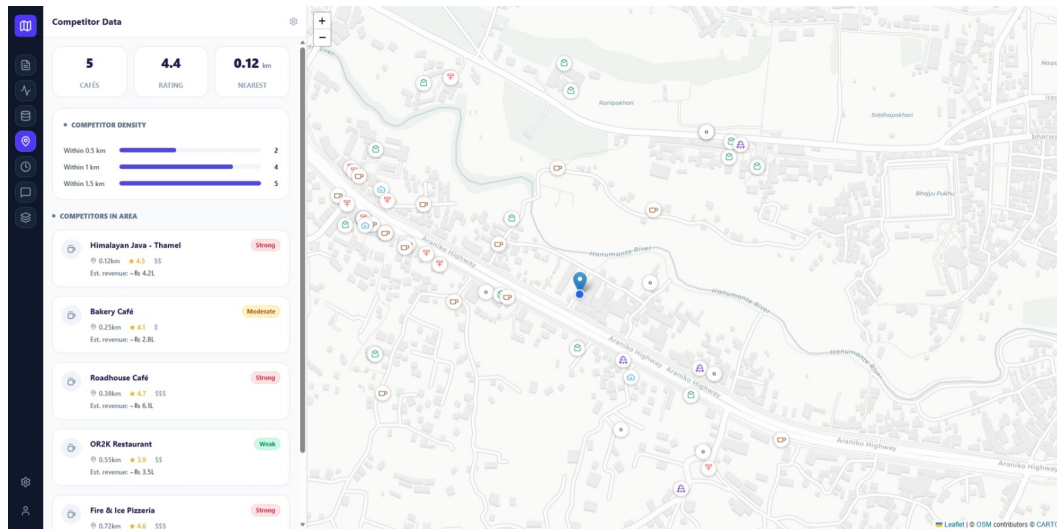


Figure 6.7: Competitor Data view. Provides an analysis of market saturation and details of nearby competing businesses.

6.4.4 Foot Traffic Analysis

The *Foot Traffic Analysis* (Figure 6.9) provides temporal patterns to help assess the location's liveliness at different times:

- **Activity Patterns:** Bar charts visualize hourly activity and weekly patterns, indicating the busiest times and days.
- **Peak Periods:** Identifies the percentage of traffic during Morning Rush, Lunch Peak, and Evening Social hours.

6.4.5 Area Reviews

Finally, the *Area Reviews* tab (Figure 6.10) captures the qualitative sentiment of the selected location:

- **Sentiment Analysis:** A progress bar format indicating the proportion of positive, neutral, and negative reviews in the area.
- **Top Keywords:** Highlights frequently mentioned terms (e.g., “good coffee”, “crowded”, “expensive”) for quick insights.
- **Recent Reviews:** Displays snippets of actual reviews to provide deeper qualitative context.

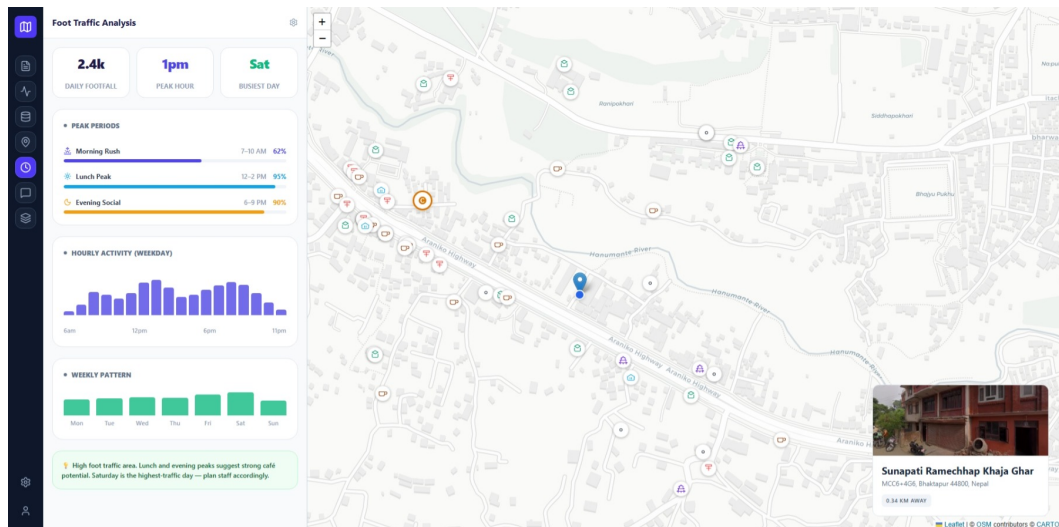


Figure 6.8: Foot Traffic Analysis view. Visualizes temporal activity trends, peak hours, and daily footfall metrics.

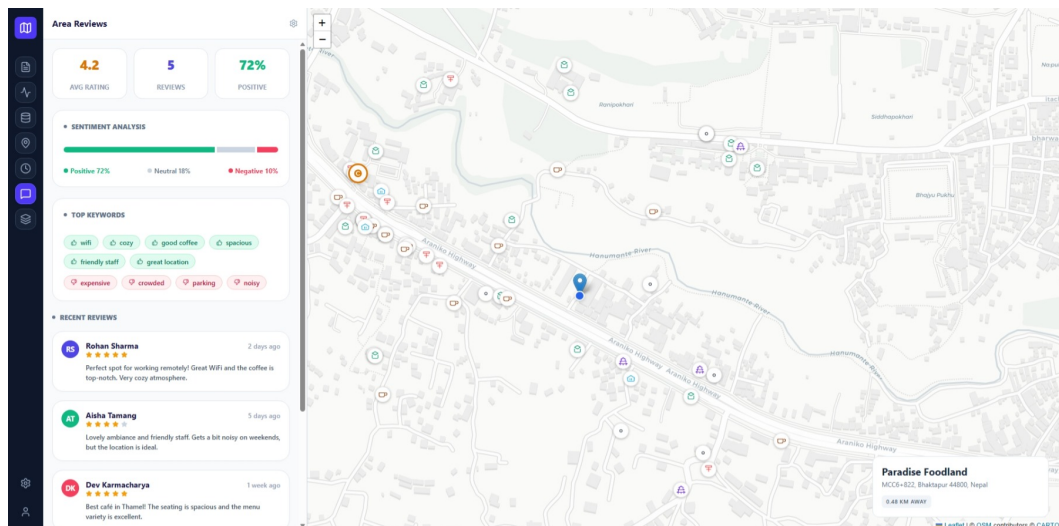


Figure 6.9: Area Reviews view. Summarizes local customer sentiment and lists top keywords and recent reviews.

Chapter 7

Conclusion and Future Enhancements

7.1 Conclusion

SiteX has been successfully developed as a comprehensive Location Intelligence platform, tailored for cafe and restaurant entrepreneurs to mitigate financial risks associated with site selection.

The platform's core achievement is its advanced Heterogeneous Graph Neural Network (HeteroGNN) architecture, leveraging PyTorch Geometric, SpatiaLite, and OSMnx. By modeling the urban environment as an interconnected graph, the system accurately captures complex spatial dependencies—such as competitor proximity and POI accessibility—surpassing traditional distance-based methods.

Furthermore, the interactive Intake Wizard enables personalized analytics through a Business Requirements Profile. SiteX computes a dynamically contextualized suitability score that adjusts the base GNN prediction according to business scale, target demographic, pricing, and competition. Coupled with a rich dashboard offering insights like Foot Traffic Analysis and Area Reviews, the system empowers users to make highly informed, data-driven real estate decisions in the Kathmandu Valley.

7.2 Future Enhancements

While the current iteration of SiteX provides a robust and reliable foundation for site evaluation, there are several avenues for future enhancement to further elevate its accuracy, scalability, and feature set:

- **Real-Time Mobility and Traffic Integration:** Currently, foot traffic is estimated based on static proxies and scraped POI data. Integrating live mobility feeds from telecom providers or mobile location APIs would allow for dynamic, time-of-day specific footfall assessments.
- **To-Let Property Listings and API Integration:** While the current system focuses on scoring user-selected map coordinates, a valuable future enhancement is

the integration of real commercial "To-Let" listings. Integrating with major real estate platforms via APIs would allow the system to continuously ingest, score, and recommend live rental properties as they hit the market, bridging the gap between analysis and immediate action.

- **Advanced Explainable AI (XAI) Visualizations:** While the current dashboard provides strong contextual breakdowns, future versions could project the HeteroGNN's attention weights directly onto the interactive map, visually highlighting exactly which neighboring streets or competitors most heavily influenced the score.
- **Temporal Graph Modeling:** Urban landscapes evolve. Upgrading the architecture to a Spatio-Temporal Graph Neural Network (STGNN) would allow the model to account for seasonal variations, gentrification trends, and long-term urban development patterns.
- **Geographic Expansion:** The data pipelines and graph construction logic have been built to be highly modular. A primary next step is scaling the system beyond the Kathmandu and Bhaktapur regions to encompass other major economic hubs across Nepal.

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